

Shakir Ahmed

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SKILLS

- Programming Languages:**
Python (pandas, NumPy, SciPy), C++,
SQL, Bash.
- ML:**
Scikit-Learn, TensorFlow, Keras,
PyTorch.
- Data & Backend:**
FastAPI, PostgreSQL, Microsoft Azure,
AWS.
- MLOps & Infrastructure:**
DVC, MLflow, Jenkins, Prefect, Docker,
Git, CI/CD, Linux.

LANGUAGES



PROJECTS

Geometry Aware Fourier Neural Network
for Acoustic Dynamics

AWARDS

Silver Medal Winner
The 2016 University Physics Competition,
Canada

CERTIFICATION

Deep Learning Specialization *DeepLearning.AI*

Advanced Machine Learning
and Signal Processing *IBM*

8th Workshop on Collaborative
Scientific Software Development
and Management of Open Source
Scientific Packages *ICTP*

ABOUT ME

Master of Science candidate in Physics specializing in
Computational Modeling, Quantum Physics, and Machine Learning.
Seeking to utilize expertise in data analysis, machine learning, and
analytical rigor to solve complex problems in a dynamic and
innovative environment.

PROFESSIONAL EXPERIENCE

- Working Student - Algorithms**
R&D Software April. '24 - March '26
Luna Innovations Germany GmbH, Cologne, Germany
- Architected and deployed full data preparation, training, evaluation, and
prediction pipelines for industrial deep learning models utilizing PyTorch,
TensorFlow, and DVC.
 - Engineered automated data processing pipelines from scratch, parsing and
filtering massive physical HDF5 datasets for high-precision temperature
and acoustic signal analysis.
 - Developed and automated their image annotation workflow for the
primary industrial model.
 - Developed and deployed internal application utilizing FastAPI and
PostgreSQL, building highly functional, database-driven microservice
architectures.

- Subject Matter Expert**
Academics Sep. '21 - Dec. 2022
Shikho Technologies Bangladesh Ltd, Dhaka, Bangladesh
- Translated highly abstract technical physics concepts into structured,
accessible presentation frameworks.
 - Developed 65+ educational scripts, collaborating with design teams to
ensure technical accuracy and rigorous standards.

- System Administrator**
Hardware March '17 - Sep. 2017
HEXit, Sylhet, Bangladesh
- Administered and secured 7+ Linux (CentOS) web servers,
monitoring logs and mitigating system vulnerabilities to prevent
cyber intrusions.
 - Designed and implemented network DNS map.

EDUCATION

- Master of Science in Physics**
University of Cologne 2023 - Present
- Bachelor of Science in Physics**
Shahjalal University of Science and Technology 2015 - 2019
Project: Wigner Monte Carlo